

PROGRESS UPDATE · JUNE 2026

NepError: A Hierarchical Diagnostic Taxonomy of Failure Modes in Nepali NLI

Pema Tshering Sherpa

Nepali NLP Evaluates by Score — Not by Diagnosis

- NLUE 2025 (KU ILPRL) establishes the first public Nepali NLI benchmarks — MNLI-Nepali, RTE-Nepali, XNLI-Nepali
- All current evaluations report aggregate accuracy/F1 only — no framework exists to categorize *why* models fail
- NepError proposes the first cross-model failure taxonomy: 4 tiers, 8 codes, 4 models, 3 datasets

Current State

XLNet achieves 80.8% F1 on MNLI-Nepali — no diagnostic breakdown of the remaining 19.2% of failures.

NepError Asks

What specific linguistic property caused each failure instance? Enables targeted model and corpus improvement.

Eight Failure Codes Organized Across Four Linguistic Tiers

Tier	Name	Codes	Key Signal
T1	Surface / Input	T1a Tokenization Fragmentation, T1b Romanization Noise	Devanagari morpheme split; Roman script OOV
T2	Morpho-Syntactic	T2a Negation Particle Drop, T2b Honorific-Verb Mismatch	छैन/होइन ignored; हजुर register confused
T3	Socio-Pragmatic	T3a Code-Switching (Neplish), T3b Domain Vocabulary Gap	English in Devanagari sentence; legal/agri/forestry terms
T4	Ambiguous / Noise	T4a Genuine Ambiguity, T4b Translation Artifact	Native speakers disagree; machine-translated sentence is broken



Annotators apply codes using a strict 8-step decision checklist. One code per instance.

Four Models, Same Data — Four Distinct Failure Personalities

Architecture-Level Findings

- **CHiPSAL-BERT** underperforms multilingual models despite Nepali-specific pretraining — 74% vs. 80.8%. Hypothesis: domain gap between news-text pretraining and machine-translated NLI pairs.
- **Distinct dominant failure modes on MNLI-Nepali:** XLM-R conflates contradiction with entailment (118 cases); mDeBERTa and CHiPSAL retreat to neutral on contradictions (132, 142 cases); Llama overclaims entailment zero-shot (36 cases).
- **Negation markers** appear in ~23–25% of failures across all four models. Consistent across architectures — suggests dataset-level difficulty. Corpus base rate comparison pending (Jinu's suggestion).

Corpus-Level Findings

- **Dataset cartography:** 69% of MNLI-Nepali training instances fall in the ambiguous region. English NLI benchmarks typically show 40–50%. Likely evidence of translation noise — native speaker verification planned.
- **16–20 pp accuracy drop** from MNLI to RTE across all models. Expected to surface T3b (domain vocabulary) as dominant failure mode on RTE.

Inference Complete — Moving into Annotation Phase

Task	Status	Owner
MNLI-Nepali inference (all 4 models)	Complete	Pema
RTE-Nepali inference (XLM-R, mDeBERTa)	Complete	Pema
XNLI-Nepali inference	Queued — next run	Pema
Annotator guidelines document	Complete	Pema
Negation base-rate regex on full corpus	To do (pre-annotation)	Pema
Pilot annotation (50 rows, $\kappa \geq 0.70$)	Ready to start	Pema + KU co-annotator
Native speaker check on ambiguous-region sample	Planned	KU team
GitHub repo shared with ILPRL	This week	Pema



Open question for the team: How should we frame the 69% ambiguous finding — translation noise vs. genuine Nepali task difficulty? Jinu's suggestion of a native speaker sample is the proposed resolution.